Homework 6- Due Date April 22, 11:59 P.M.

Problem 1. Let X and Y be two random variables, whose joint PMF $p_{X,Y}(x,y)$ is given by the following table:

$x=1$	$\frac{3}{16}$	$\frac{1}{16}$	$\frac{1}{8}$
$x=0$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{4}$
	$y=-1$	$y=0$	$y=1$

Using this table, evaluate the following quantities.

(1.1) $Var(X|Y=0)$.

(1.2) $p_{X|Y}(1|A)$, where $A = \{Y^2 > 0\}$

Problem 2. Consider a joint probability density function $f(x,y)$ is given by

$$\begin{cases} \frac{16y}{x^3} & x > 2, 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

(i) Find $E[X]$ and $E[Y]$.

(ii) Find $Cov(X,Y)$

(iii) Are X and Y correlated? Why/Whynot?

Problem 3. A book has n typos. Two proofreaders, Sarah and Hannah, independently read the book. Sarah catches each typo with probability p_1 and misses it with probability $1-p_1$, independently, and likewise for Hannah, who has probabilities p_2 of catching and $1-p_2$ of missing each typo. Let X_1 be the number of typos caught by Sarah, X_2 be the number caught by Hannah, and X be the number caught by at least one of the two proofreaders.

(a) Find the distribution of X.

(b) For this part only, assume that $p_1 = p_2$. Find the conditional distribution of X_1 given that $X_1 + X_2 = t$

Problem 4. Let X be the number of Heads in 10 fair coin tosses.

(a) Find the conditional PMF of X, given that the first two tosses both land Heads.

(b) Find the conditional PMF of X, given that at least two tosses land Heads.

Problem 5. Let X and Y be independent random variables following Poisson distributions with parameters λ_1 and λ_2. Find the distribution of $Z = X + Y$.

Problem 6. Each of $n \geq 2$ people puts his or her name on a slip of paper (no two have the same name). The slips of paper are shuffled in a hat, and then each person draws one (uniformly at random at each stage, without replacement). Find the mean and standard deviation of the number of people who draw their own names.